

# Renfeng Peng

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🌐 <http://jimmpeng1998.github.io/>

🏠 Sept. 30th, 1998, Changde, Hunan

🔍 Renfeng Peng at Google scholar

📍 Workstation 7, Room 215, Innovation Centre, 72 Tat Chee Avenue, Kowloon Tong, Kowloon, Hong Kong



## 👤 Employment

2026.07 – Present

### 📖 Research Assistant

Department of Data Science, City University of Hong Kong, Hong Kong SAR

Host: Prof. Fenglei Fan

## 🎓 Education

2021.09 – 2026.06

### 📖 Ph.D. in computational mathematics

Academy of Mathematics and Systems Science, Chinese Academy of Sciences; and University of Chinese Academy of Sciences, Beijing, China

Supervisor: Prof. Ya-xiang Yuan

2017.09 – 2021.07

### 📖 Bachelor in mathematics and applied mathematics

School of Mathematical Sciences, Tongji University, Shanghai, China.

## 📄 Publications

### Refereed papers

- 1 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Low-rank optimization on Tucker tensor varieties”. In: *Mathematical Programming* 214 (2025), pp. 357–407. 🔗 DOI: 10.1007/s10107-024-02186-w.
- 2 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Optimization on product manifolds under a preconditioned metric”. In: *SIAM Journal on Matrix Analysis and Applications* 46.3 (2025), pp. 1816–1845. 🔗 DOI: <https://epubs.siam.org/doi/abs/10.1137/24M1643773>.
- 3 Chengkai Zhu, **Renfeng Peng**, Bin Gao, and Xin Wang. “Riemannian Optimization for Holevo Capacity”. In: *2025 IEEE International Symposium on Information Theory (ISIT)*. 2025, pp. 1–6. 🔗 DOI: 10.1109/ISIT63088.2025.11195337.
- 4 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Riemannian preconditioned algorithms for tensor completion via tensor ring decomposition”. In: *Computational Optimization and Applications* 88 (2024), pp. 443–468. 🔗 DOI: 10.1007/s10589-024-00559-7.
- 5 Ling-Xi Zhang, **Ren-Feng Peng**, and Jun-Feng Yin. “A second order numerical scheme for fractional option pricing models”. In: *East Asian J. Appl. Math* 11 (2021), pp. 326–348. 🔗 DOI: 10.4208/eajam.020820.121120.

### Preprints

- 1 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. “Quotient geometry of tensor ring decomposition”. In: *ArXiv preprint arXiv:2601.21874* (2026).
- 2 Pengfei Hao, **Renfeng Peng**, Yang Kuang, and Bin Gao. “High-dimensional extreme eigenvalue problems: low-rank tensor parametrization and optimization”. In: *ArXiv preprint arXiv:2609.30138* (2026).

- 3 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. "First-order methods on bounded-rank tensors converging to stationary points". In: *ArXiv preprint arXiv:2503.04523* (2025).
- 4 **Renfeng Peng**, Chengkai Zhu, Bin Gao, Xin Wang, and Ya-xiang Yuan. "Normalized tensor train decomposition". In: *ArXiv preprint arXiv:2511.04369* (2025).
- 5 Bin Gao, **Renfeng Peng**, and Ya-xiang Yuan. "Desingularization of bounded-rank tensor sets". In: *ArXiv preprint arXiv:2411.14093* (2024).

## Research Interests and Skills

- Research  Tensor optimization, manifold optimization, and machine learning.
- Languages  Mandarin Chinese, English.
- Coding  Matlab, C/C++,  $\text{\LaTeX}$ , ...

## Awards and achievements

- 2026/07  CAS President's Excellent Award (5,000 CNY).
- 2025/09  AMSS President's Special Award (32,000 CNY).
- 2024/09  Loo-Keng Hua Scholarship.
- 2021/09  Loo-Keng Hua Scholarship.

## Presentations

- 2026/04  2026 International Conference on Optimization: Techniques and Applications (ICOTA 2026). Shanghai, China.
- 2025/08  The 15th International Conference on Numerical Optimization and Numerical Linear Algebra. Harbin, China.
- 2025/07  The 8th International Conference on Continuous Optimization. Los Angeles, U.S.
- 2024/11  The 7th PKU PhD Symposium for Applied Mathematics. Beijing, China.

## Referee for journals

- Mathematical Programming Computation
- Optimization Methods and Software